

ROOT BOUNDS FOR CHAIN POLYNOMIALS BEYOND THE COHEN–MACAULAY WORLD

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ABSTRACT. The chain polynomial $c_P(t) = \sum_{k \geq 0} c_k(P) t^k$ of a finite poset counts its chains by size. We prove that for every finite graded poset all real roots of $c_P(t)$ lie in $[-4, 0]$. Graded posets need not be Cohen–Macaulay, so the h -vector route to such bounds is unavailable here.

The proof rests on a sign condition. Let A be the 0/1 matrix of the strict order relation, so that $(I + A)^{-1}$ is the Möbius function and $P_{xy}(z) = \sum_k z^k (A^k)_{xy}$ counts chains from x to y by length; the chain polynomial of the proper part is $P_{\hat{0}\hat{1}}(t)/t$. If the entries of $(I + aA)^{-1}$ all carry the sign $(-1)^{\sigma(x)+\sigma(y)}$ for some 2-coloring σ of the elements, then every P_{xy} has all its real roots in $[-a, 0]$. A local condition on the graph, checked one pair of vertices at a time, certifies that sign pattern.

Larger rank-2 intervals give shorter intervals of roots. For the subspace lattice $L_n(\mathbb{F}_q)$ we obtain $[-4/(q+1), 0]$, a proper subinterval of the $[-1, 0]$ known for geometric lattices once $q \geq 4$. The method also applies to relations that are not partial orders: for every finite group, the polynomial counting chains of iterated proper normal subgroups has all real roots in $[-4, 0]$. Root location has a probabilistic reading, since $-c_{\bar{P}}(-p)$ is the expected reduced Euler characteristic of the order complex of a random induced subposet at density p .

1. INTRODUCTION

The chain polynomial $c_L(t) = \sum_{k \geq 0} c_k(L) t^k$ of a finite poset L counts chains of each size. This paper replaces the usual topological input for root bounds—shellability, Cohen–Macaulayness, h -vectors—with a sign condition on a matrix inverse: a sign-alternation (“Möbius alternation”) condition on $(I + A)^{-1}$, read off a directed graph rather than an order complex.

This sits alongside an active real-rootedness program. Athanasiadis–Kalampoglia–Evangelinou [AK23] conjectured that $c_L(t)$ is real-rooted for every finite geometric lattice, and verified it for the partition lattices and their type B analogues; further families were settled by Brändén–Saud Maia Leite [BS25; BS26], who establish full real-rootedness, with all zeros in $[-1, 0]$, for perfect matroid designs, Dowling lattices, and the flats of paving matroids (the general case remains open). Related results for noncrossing partition lattices and rank-selected Cohen–Macaulay simplicial posets appear in Athanasiadis–Douvropoulos–Kalampoglia–Evangelinou [ADK24]. Real-rootedness is the stronger property; this paper proves the weaker confinement

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of the *real* roots alone, but under hypotheses weak enough to leave the simplicial world behind.

Where the order complex is well behaved, root bounds already follow from homological data: geometric and distributive lattices are shellable [Bjö80], hence Cohen–Macaulay, and the classical f/h substitution confines all real roots of $c_L(t)$ to $[-1, 0]$ (§6.4).

The sign-alternation condition. For a weighted DAG G with adjacency matrix A (nilpotent, since G is acyclic), the matrix

$$P_G(-1) = (I + A)^{-1} = \sum_{k \geq 0} (-1)^k A^k$$

generalizes the classical Möbius function: on the order DAG of a finite poset, $(P_G(-1))_{xy} = \mu(x, y)$. We introduce a *sign-alternation condition* on $P_G(-1)$ —requiring its entries to have signs prescribed by a parity function $\sigma : V \rightarrow \mathbb{Z}_2$ —and prove (Theorem 3.2) that this condition alone forces every walk generating polynomial $P_{G,xy}(z)$ to have all real roots in $[-a, 0]$, provided the edge weights are nonnegative.

Section 5 gives a graph-theoretic certificate for the sign condition: *stitchability*, a local domination condition on edge pairs. In quantitative form (Theorem 5.9), measuring the pooled weight of each same-parity edge’s opposite-parity relays by ratios λ_o, λ_e , the root interval becomes $[-(\lambda_o^{-1/2} + \lambda_e^{-1/2})^2, 0]$, recovering $[-4, 0]$ at $\lambda_o = \lambda_e = 1$ and shrinking as relay capacity grows. This is sharp when $\lambda_o = \lambda_e$ and in the one-sided case; for general asymmetric pairs sharpness is supported by a transfer-matrix analysis and by computation, but not proved (Remark 5.10).

Applications and new results. Beyond the Cohen–Macaulay world we obtain:

- $[-4, 0]$ for the chain polynomial of every finite graded poset, with no minimum or maximum needed (Corollary 6.15);
- $[-4/c, 0]$ when every rank-2 interval has at least c middle elements (Corollary 6.17), hence $[-4/(q+1), 0]$ for the subspace lattice $L_n(\mathbb{F}_q)$ (Theorem 6.18), a proper subinterval of $[-1, 0]$ once $q \geq 4$;
- $[-1, 0]$ for every finite Eulerian poset (§6.2), where the Euler relation $\mu(x, y) = (-1)^{\rho(y) - \rho(x)}$ is verbatim our sign hypothesis, and $[-1/k, 0]$ for the k -Eulerian posets of Bayer and Heteyi [BH02];
- $[-4, 0]$ for the f -polynomial of the barycentric subdivision of a pure simplicial complex, and $[-2, 0]$ for closed pseudomanifolds (§6.4).

Proposition 6.8 appears to be new, real-rootedness there being open even for face lattices of polytopes.

The reach extends past posets to objects with no order complex (Section 7): for any two finite groups $G_0 \supseteq H_0$, all real roots of the polynomial counting chains of iterated proper normal subgroups from G_0 to H_0 lie in $[-4, 0]$, the proof using nothing beyond the Jordan–Hölder theorem. Section 8 closes with the coprime relation on the integers, an open case that motivated the framework.

In the classical settings, the same condition recovers the known $[-1, 0]$ bounds for geometric and distributive lattices (Section 6): Rota’s NBC theorem [Rot64] and the classical Möbius formula supply the sign patterns, and Theorem 3.2 converts them directly into root bounds. These bounds are not new; the point is that a single sign computation, elementary and self-contained, reproduces them.

Definitions. Let G be a directed acyclic graph (DAG) on n vertices with edge weights in $\mathbb{R}_{>0}$; the adjacency matrix $A \in \mathbb{R}^{n \times n}$ has A_{ij} the weight of the edge $i \rightarrow j$, or 0 if there is none. Since G is acyclic, A is nilpotent; let N be least with $A^{N+1} = 0$.

The *walk generating polynomial* of G is the matrix-valued polynomial

$$(1.1) \quad P_G(z) = \sum_{k=0}^N z^k A^k = (I - zA)^{-1} \in \mathbb{R}[z]^{n \times n}.$$

The (x, y) entry $P_G(z)_{xy}$ is a polynomial in z whose coefficient of z^k is the total weight of all length- k walks from x to y (weight of a walk = product of edge weights). When all edge weights are nonnegative, every coefficient of $P_G(z)_{xy}$ is nonnegative, so all real roots of $P_G(z)_{xy}$ are nonpositive.

Chain Polynomials of Posets. Let $(P, \leq)$ be a finite partially ordered set. A *chain* in P is a totally ordered subset; write $|C|$ for its size. The *chain polynomial* of P is

$$(1.2) \quad c_P(t) := \sum_{k \geq 0} c_k(P) t^k,$$

where $c_k(P)$ counts chains of size k (including the empty chain, so $c_0(P) = 1$). Chain polynomials have been studied extensively in recent work on real-rootedness [AK23; ADK24; BS25; BS26].

For any finite poset P with a unique minimum $\hat{0}$ and maximum $\hat{1}$, define the *order DAG* of P to be the DAG with vertex set P and a directed edge from x to y (weight 1) whenever $x < y$ in P . (Edges run between all comparable pairs, not merely covering relations.) A walk of length k from $\hat{0}$ to $\hat{1}$ in this DAG visits $k+1$ elements $\hat{0} = v_0 < v_1 < \dots < v_k = \hat{1}$, which is precisely a chain of size $k+1$ containing both endpoints.

Proposition 1.1 (Chain–Walk Equivalence; proved in Section 6). *For any finite poset L with a unique minimum $\hat{0}$ and a unique maximum $\hat{1}$, where $\hat{0} \neq \hat{1}$,*

$$(1.3) \quad c_L(t) = \frac{(t+1)^2}{t} P_{L, \hat{0}, \hat{1}}(t).$$

(Since $P_{L, \hat{0}, \hat{1}}(0) = 0$, we have $t \mid P_{L, \hat{0}, \hat{1}}(t)$ and the right-hand side is a polynomial.) Equivalently, writing $\bar{L} = L \setminus \{\hat{0}, \hat{1}\}$ for the proper part,

$$c_L(t) = (1+t)^2 c_{\bar{L}}(t), \quad P_{L, \hat{0}, \hat{1}}(t) = t c_{\bar{L}}(t).$$

In particular, $t = -1$ is always a root of $c_L(t)$ of multiplicity at least two, every real root of $c_L(t)$ other than $t = -1$ is a nonzero real root of $P_{L, \hat{0}, \hat{1}}(t)$, and every nonzero real root of $P_{L, \hat{0}, \hat{1}}(t)$ is a real root of $c_L(t)$.

For a DAG G and $a > 0$, write aG for the DAG with each edge weight scaled by a (adjacency matrix aA). Then

$$(1.4) \quad P_G(-a) = \sum_{k=0}^N (-a)^k A^k = \sum_{k=0}^N (-1)^k (aA)^k = P_{aG}(-1);$$

Organization. Sections 4 and 5 develop graph sums and stitchability; Section 6 applies them to chain polynomials, Section 7 to normal-subgroup chains, and Section 8 states the open coprime case.

2. SIGN-ALTERNATING MATRICES

Definition 2.1. Let $\sigma : \{1, \dots, n\} \rightarrow \mathbb{Z}_2$ be a *parity function*. A matrix $M \in \mathbb{R}^{n \times n}$ has sign σ , written $M \in \text{ALT}_\sigma$, if

$$(2.1) \quad M_{ij} = (-1)^{\sigma(i)-\sigma(j)} |M_{ij}| \quad \text{for all } i, j.$$

Equivalently, $M_{ij} \geq 0$ when $\sigma(i) = \sigma(j)$ and $M_{ij} \leq 0$ when $\sigma(i) \neq \sigma(j)$. A zero entry satisfies both, so the condition constrains M_{ij} only where $M_{ij} \neq 0$. This matters below: in a DAG $P_{G,xy} \equiv 0$ whenever y is unreachable from x , and such pairs impose nothing.

Lemma 2.2 (Closure under products and sums). *If $M, N \in \text{ALT}_\sigma$, then $M + N \in \text{ALT}_\sigma$ and $MN \in \text{ALT}_\sigma$. Any nonnegative scalar multiple of a sign- σ matrix also has sign σ .*

Proof.

$$(M+N)_{ij} = (-1)^{\sigma(i)-\sigma(j)} (|M_{ij}| + |N_{ij}|), \quad (MN)_{ij} = \sum_k M_{ik} N_{kj} = (-1)^{\sigma(i)-\sigma(j)} \sum_k |M_{ik}| |N_{kj}|,$$

$$\text{and } (cM)_{ij} = (-1)^{\sigma(i)-\sigma(j)} c |M_{ij}| \text{ for } c \geq 0. \quad \square$$

3. WALK GENERATING POLYNOMIALS AND ROOT BOUNDS

Derivative Formula and Sign Preservation.

Theorem 3.1. *Let A be a strictly upper triangular matrix (for instance, the adjacency matrix of a DAG under a topological vertex ordering), and let $f(z) = (I - zA)^{-1}$. If $f(1) \in \text{ALT}_\sigma$, then $f(z) \in \text{ALT}_\sigma$ for all $z \geq 1$, and in particular $f(z)_{ij} \neq 0$ for $z > 1$ whenever $f(z)_{ij} \neq 0$.*

Proof. From $(I - A)f(1) = I$ we get $Af(1) = f(1) - I$; and $(A^k)_{ii} = 0$ for $k \geq 1$ gives $f(1)_{ii} = 1$, so $f(1) - I$ has zero diagonal and agrees with $f(1)$ off it. Hence $f(1) \in \text{ALT}_\sigma \Rightarrow f(1) - I \in \text{ALT}_\sigma$.

Induction on m , using $\frac{d}{dz} M^{-1} = -M^{-1} M' M^{-1}$ with $M = I - zA$, gives

$$f^{(m)}(z) = m! A^m (I - zA)^{-(m+1)} = m! [Af(z)]^m f(z), \quad \text{so} \quad f^{(m)}(1) = m! [f(1) - I]^m f(1) \in \text{ALT}_\sigma$$

for every $m \geq 0$, by Lemma 2.2.

Since A is nilpotent the Taylor expansion at $z = 1$ is finite:

$$f(z) = \sum_{m=0}^N \frac{(z-1)^m}{m!} f^{(m)}(1), \quad |f(z)_{ij}| = \sum_{m=0}^N \frac{(z-1)^m}{m!} |f^{(m)}(1)_{ij}| \quad (z \geq 1).$$

For $z \geq 1$ the coefficients are nonnegative, so $f(z) \in \text{ALT}_\sigma$; for $z > 1$ they are strictly positive, so $f(z)_{ij} \neq 0$ whenever $f(z)_{ij} \neq 0$. $\square$

Root Bounding via Scaling. Theorem 3.1 excludes roots above 1 for any strictly upper triangular A . Applying it to the scaled adjacency matrix $-aA$ gives the general root bound.

Theorem 3.2 (Root Bounding). *Let G be a DAG with nonnegative edge weights. If $P_G(-a) = (I + aA)^{-1}$ has sign σ for some $a > 0$, then for every x, y with $P_{G,xy} \neq 0$ all real roots of $P_{G,xy}(z)$ lie in $[-a, 0]$.*

Proof. All coefficients of $P_{G,xy}(z) = \sum_k z^k (A^k)_{xy}$ are nonnegative, so there are no positive roots. Put $B = -aA$ and $h(z) = (I - zB)^{-1}$, so that $P_{G,xy}(-az) = h(z)_{xy}$ and $h(1) = (I + aA)^{-1} \in \text{ALT}_\sigma$. Theorem 3.1 gives $h(z)_{xy} \neq 0$ for $z > 1$, i.e. no root of $P_{G,xy}$ in $(-\infty, -a)$. $\square$

Every root bound below is an instance of this theorem: the sign pattern comes either from a classical Möbius formula or from stitchability (Section 5).

Proposition 3.3 (Every derivative is localized). *Under the hypotheses of Theorem 3.2, for every x, y and every $m \geq 0$:*

- (i) $(-1)^{m+\sigma(x)-\sigma(y)} P_{G,xy}^{(m)}(-a) \geq 0$, the inequality being strict exactly when $P_{G,xy}^{(m)}(-a) \neq 0$; that is, the derivatives at the left endpoint are sign-definite, alternating with m ; and
- (ii) for every m with $P_{G,xy}^{(m)} \neq 0$, all real roots of $P_{G,xy}^{(m)}$ lie in $[-a, 0]$.

Proof. Put $B = -aA$ and $h(z) = (I - zB)^{-1} = (I + zaA)^{-1}$, so that $P_{G,xy}(-az) = h(z)_{xy}$ and $h(1) = P_G(-a)$ has sign σ . The derivative formula of Theorem 3.1 gives

$$h^{(m)}(z) = m! B^m (I - zB)^{-(m+1)} = m! [Bh(z)]^m h(z), \quad Bh(z) = \frac{1}{z}(h(z) - I),$$

the last identity from $(I - zB)h(z) = I$. For $z \geq 1$, Theorem 3.1 gives $h(z) \in \text{ALT}_\sigma$; since $h(z) - I$ has zero diagonal and the same off-diagonal entries it too lies in ALT_σ , and $\frac{1}{z} > 0$, so $Bh(z) \in \text{ALT}_\sigma$. By Lemma 2.2, $h^{(m)}(z) = m! [Bh(z)]^m h(z) \in \text{ALT}_\sigma$ for all $z \geq 1$ and all $m \geq 0$.

Differentiating $P_{G,xy}(-az) = h(z)_{xy}$ in z yields $(-a)^m P_{G,xy}^{(m)}(-az) = h^{(m)}(z)_{xy}$. At $z = 1$, $(-a)^m P_{G,xy}^{(m)}(-a) = h^{(m)}(1)_{xy}$, which carries the sign $(-1)^{\sigma(x)-\sigma(y)}$ weakly; dividing by $(-a)^m$ (sign $(-1)^m$) gives (i). The value may well vanish even when $P_{G,xy}^{(m)}$ does not: for the order DAG of the four-element chain $x < a < b < y$ one has $P_{x,y}(z) = z(1+z)^2$, so $P_{x,y}$ and $P'_{x,y}$ both vanish at $a = 1$. For (ii): differentiating the nonnegative-coefficient polynomial $P_{G,xy}$ keeps the coefficients nonnegative, so $P_{G,xy}^{(m)}$ has no positive roots. And by the Taylor-positivity argument of Theorem 3.1—the coefficients $h^{(m+r)}(1)_{xy}$ all share the sign $(-1)^{\sigma(x)-\sigma(y)}$, so $h^{(m)}(z)_{xy} = \sum_{r \geq 0} \frac{(z-1)^r}{r!} h^{(m+r)}(1)_{xy} \neq 0$ for $z > 1$ whenever $P_{G,xy}^{(m)} \neq 0$ —the derivative has no root in $(-\infty, -a)$. Hence all real roots lie in $[-a, 0]$. $\square$

Remark 3.4 (Not Rolle, not Gauss–Lucas). For a real-rooted $P_{G,xy}$ part (ii) would follow from Rolle or Gauss–Lucas. But real-rootedness can fail here, and then the convex hull of the roots spills outside $[-a, 0]$; Proposition 3.3 instead reads the bound off the sign-definiteness of $A^m(I + aA)^{-(m+1)}$, with no real-rootedness input. The gap is not hypothetical. For a geometric lattice, real-rootedness of c_L is the conjecture of [AK23] and remains open, so neither Rolle nor Gauss–Lucas is available; yet Proposition 3.3 applies at $a = 1$ by Remark 6.3, placing all real roots of every derivative of $P_{L, \hat{0}, \hat{1}}$ in $[-1, 0]$ unconditionally. Note that these are derivatives of the walk generating polynomial; since $P_{L, \hat{0}, \hat{1}}(t) = t c_L(t)$, the corresponding statement for the derivatives of c_L does not follow, and we do not know whether it holds.

Remark 3.5 (Monotonicity in the edge weight). The same computation gives monotonicity in the scaling parameter: for $a' \geq a$, $\frac{d^m}{da'^m} P_{G,xy}(-a') = (-1)^m P_{G,xy}^{(m)}(-a')$

has constant sign $(-1)^{\sigma(x)-\sigma(y)}$ in m by part (i), so $(-1)^{\sigma(x)-\sigma(y)}P_{G,xy}(-a')$ is absolutely monotone on $[a, \infty)$.

The condition $P_G(-a) \in \text{ALT}_\sigma$ is equivalent to saying the scaled graph aG *Möbius-alternates*: $P_{aG}(-1) = (I + aA)^{-1}$ has sign σ . The next sections develop checkable conditions for this.

4. GRAPH SUMS AND ALTERNATION PRESERVATION

Let G_1 and G_2 be weighted DAGs on the same vertex set with adjacency matrices A and B . Define the *graph sum* $G_1 \oplus G_2$ to be the DAG with adjacency matrix $A + B$, assuming this is still acyclic (equivalently, $C := A + B$ is nilpotent).

Proposition 4.1. *Let A, B be strictly upper triangular matrices (under a common vertex ordering) with $C = A + B$ also nilpotent. If $(I - A)^{-1}$ and $(I - B)^{-1}$ each have sign σ , then $(I - C)^{-1}$ has sign σ .*

Proof. Put $P_A = (I - A)^{-1} - I$ and $P_B = (I - B)^{-1} - I$. As in Theorem 3.1, both have zero diagonal and agree off it with $(I - A)^{-1}$, $(I - B)^{-1}$, so $P_A, P_B \in \text{ALT}_\sigma$.

Since C is nilpotent, $(I - C)^{-1} = \sum_{n \geq 0} (A + B)^n$, and every monomial in $(A + B)^n$ is a word in A and B . Grouping words by their first letter and their number of letter-switches — a word with i switches beginning on A is $A^{a_0} B^{b_0} A^{a_1} \dots$ with all exponents ≥ 1 , and these sum to $P_A P_B P_A \dots$ ($i + 1$ alternating factors) — each word is counted exactly once, so

$$(4.1) \quad (I - C)^{-1} = I + \sum_{i \geq 0} \left(\underbrace{P_A P_B \dots}_{i+1} + \underbrace{P_B P_A \dots}_{i+1} \right),$$

a finite sum. The sum is finite because C is nilpotent. Since $P_A, P_B \in \text{ALT}_\sigma$, every term is a finite product of sign- σ matrices and lies in ALT_σ by Lemma 2.2; the identity I trivially has sign σ . Hence $(I - C)^{-1} \in \text{ALT}_\sigma$. $\square$

5. CONDITIONS ENSURING ALTERNATION

We develop structural criteria guaranteeing that $P_G(-1)$ has sign σ , which by Theorem 3.2 bounds all real roots of the walk generating polynomials. Throughout this section all edge weights are nonnegative.

Möbius Function and Alternation. Let G be a weighted DAG with nonnegative edge weights and adjacency matrix A . For vertices x, y define the *Möbius value*

$$(5.1) \quad \mu_G(x, y) := P_{G,xy}(-1) = \sum_{k \geq 0} (-1)^k (A^k)_{xy}.$$

When G is the order DAG of a finite poset, this recovers the classical Möbius function of the poset [Rot64; Sta11].

Finch [Fin70] develops the incidence algebra and Möbius function of an arbitrary binary relation ρ on a finite set, setting $\mu_\rho = \zeta_\rho^{-1}$ whenever ζ_ρ is invertible. For the reflexive closure of a DAG, $\zeta_\rho = I + A$ with A nilpotent, so $\det \zeta_\rho = 1$ and $\mu_\rho = (I + A)^{-1}$ is the matrix of Möbius values above.

Using $A^k = A \cdot A^{k-1}$, a direct computation gives the recursion: for $x \neq y$,

$$(5.2) \quad \mu_G(x, y) = - \sum_{x \rightarrow v} W_G(x, v) \mu_G(v, y), \quad \mu_G(x, x) = 1,$$

where the sum is over all out-neighbors v of x with edge weight $W_G(x, v)$.

We say G *Möbius-alternates* with respect to $\sigma : V \rightarrow \mathbb{Z}_2$ if

$$(5.3) \quad \mu_G(x, y) = (-1)^{\sigma(x) - \sigma(y)} |\mu_G(x, y)| \quad \text{for all } x, y.$$

Equivalently, $P_G(-1) = (I + A)^{-1}$ has sign σ . With nonnegative edge weights Theorem 3.2 applies at $a = 1$: the coefficients of $P_{G,xy}$ are nonnegative, which excludes positive roots, and the sign pattern at $z = -1$ excludes roots below -1 , leaving $[-1, 0]$. Alternation is a threshold property: once the scale is large enough the sign pattern appears, and scaling further cannot destroy it (Theorem 3.1), so the sharpest bound sits at the least such a .

Closed-form Möbius values are known for many posets and DAGs [Rot64; Sta11], making the condition directly checkable; for Eulerian posets it holds by definition (§6.2).

Remark 5.1 (Parity alignment). The colouring is essentially forced. If G Möbius-alternates and $\mu_G(x, y) \neq 0$, then comparing the sign of $P_{G,xy}$ at $z = -1$ with its sign as $z \rightarrow -\infty$, where the leading term $c_{d(x,y)} z^{d(x,y)}$ dominates, gives $d(x, y) \equiv \sigma(x) - \sigma(y) \pmod{2}$ for $d(x, y)$ the longest path length from x to y . So σ is determined up to a global flip on each connected component of the graph of pairs with $\mu_G \neq 0$. The condition is necessary but not sufficient.

Going forward, call vertices x with $\sigma(x) = 0$ *even* and those with $\sigma(x) = 1$ *odd*.

Stitchable DAGs. We now give two conditions that together guarantee Möbius alternation.

- G is **partially independent** with respect to σ if no two even vertices are adjacent (i.e., $W_G(x, y) = 0$ whenever $\sigma(x) = \sigma(y) = 0$). Since Möbius alternation with σ is equivalent to alternation with $1 - \sigma$, we may assume WLOG that there are no even-even edges.
- G is **stitchable** with respect to σ if for every same-parity pair x, y with $W_G(x, y) > 0$, there exists a vertex r of opposite parity such that

$$(5.4) \quad W_G(x, r) W_G(r, y) \geq W_G(x, y).$$

In the unit-weight case stitchability reads: between any two same-parity vertices joined by an edge lies an opposite-parity vertex. In a graded poset an intermediate rank supplies one.

Stitchability is a local *sufficient* condition, and it imposes the shortest possible relay: length two, through a single opposite-parity vertex. It is not necessary in general. Several relays can compensate collectively where no single one dominates: take $\sigma(x) = \sigma(y) = 0$ and $\sigma(r_1) = \sigma(r_2) = 1$ with $W(x, y) = 1$, $W(x, r_i) = 1$ and $W(r_i, y) = 3/4$; no individual relay dominates the direct edge, yet $\mu_G(x, y) = -1 + 2 \cdot \frac{3}{4} = \frac{1}{2} > 0$, the sign a same-parity pair requires. Remark 5.2 identifies the case in which stitchability does become necessary, and Theorem 5.9 replaces single relays by their pooled weight.

Remark 5.2 (Necessity in the transitive case). When G is the order DAG of a poset L (every comparable pair carries a direct edge), stitchability is also *necessary* for Möbius alternation under any uniform scaling. If kL Möbius-alternates with σ for some $k > 0$, then any same-parity pair (x, y) with no opposite-parity element strictly between them satisfies $\mu_{kL}(x, y) = -k \sum_{x \leq z < y} \mu_{kL}(x, z) \leq -k < 0$ by induction (every z in the sum shares the parity of x , so each term is nonnegative and the $z = x$ term equals 1), contradicting alternation. Hence, by Corollary 5.4 below,

$4L$ Möbius-alternates *if and only if* L is stitchable. For general non-transitive DAGs the condition remains sufficient but not necessary—a longer even-length relay can compensate—which is exactly what makes the coprime problem (Section 8) open.

Lemma 5.3. *If G is stitchable and partially independent with respect to σ , then G Möbius-alternates with respect to σ .*

Proof. Induct on the height $h(x)$, the length of the longest path from x . If $h(x) = 0$ then $\mu_G(x, x) = 1$ and $\mu_G(x, y) = 0$ otherwise, both of the required sign. For $h(x) = n + 1$, every out-neighbour v has $h(v) \leq n$, so $\mu_G(v, y) = (-1)^{\sigma(v)-\sigma(y)} |\mu_G(v, y)|$ and $\mu_G(x, y) = - \sum_{x \rightarrow v} W_G(x, v) \mu_G(v, y)$ by (5.2).

Case 1: x even. By partial independence every out-neighbor v of x is odd ($\sigma(v) = 1$). Substituting:

$$\begin{aligned} \mu_G(x, y) &= - \sum_{x \rightarrow v} W_G(x, v) \cdot (-1)^{1-\sigma(y)} |\mu_G(v, y)| \\ &= (-1)^{\sigma(x)-\sigma(y)} \sum_{x \rightarrow v} W_G(x, v) |\mu_G(v, y)|. \end{aligned}$$

The sum is nonnegative, so $\mu_G(x, y)$ has the required sign.

Case 2: x odd. Out-neighbors may be even or odd. Separating by parity and substituting the inductive hypothesis gives

$$(5.5) \quad \mu_G(x, y) = (-1)^{\sigma(x)-\sigma(y)} (E - O),$$

where

$$E = \sum_{\substack{x \rightarrow r \\ \sigma(r)=0}} W_G(x, r) |\mu_G(r, y)|, \quad O = \sum_{\substack{x \rightarrow v \\ \sigma(v)=1}} W_G(x, v) |\mu_G(v, y)|.$$

It remains to show $E \geq O$.

By partial independence every out-neighbor of an even vertex is odd, so Case 1 applies to each even out-neighbor r of x with $r \neq y$:

$$|\mu_G(r, y)| = \sum_{r \rightarrow v, \sigma(v)=1} W_G(r, v) |\mu_G(v, y)|.$$

(The expansion is invalid for $r = y$, where $\mu_G(y, y) = 1$; that term of E is nonnegative and we simply drop it.) Substituting and grouping by v :

$$E \geq \sum_{v: \sigma(v)=1} |\mu_G(v, y)| \sum_{\substack{r: \sigma(r)=0, r \neq y \\ x \rightarrow r \rightarrow v}} W_G(x, r) W_G(r, v).$$

For each odd v with $W_G(x, v) > 0$ and $\mu_G(v, y) \neq 0$, stitchability provides an even r with $W_G(x, r)W_G(r, v) \geq W_G(x, v)$, and this r is automatically distinct from y : if $v = y$ then $r \neq y$ by parity, while if $v \neq y$ then $\mu_G(v, y) \neq 0$ forces a walk from v to y , so the edge $y \rightarrow v$ would close a cycle and acyclicity gives $r \neq y$. Hence the inner sum is at least $W_G(x, v)$ whenever $|\mu_G(v, y)| > 0$, so $E \geq O$, and $\mu_G(x, y) = (-1)^{\sigma(x)-\sigma(y)} (E - O)$ has the correct sign. $\square$

Scaling to Ensure Alternation. Partial independence is a restrictive condition. The following removes it via a scaling and decomposition argument.

Corollary 5.4. *If G is stitchable with respect to σ , then $4G$ Möbius-alternates with respect to σ .*

Proof. Decompose $4G = R \oplus T$ by (5.6)

$$W_R(x, y) = \begin{cases} 4W_G(x, y) & \text{if } \sigma(x) = \sigma(y) = 1, \\ 2W_G(x, y) & \text{if } \sigma(x) \neq \sigma(y), \\ 0 & \text{if } \sigma(x) = \sigma(y) = 0, \end{cases} \quad W_T(x, y) = \begin{cases} 4W_G(x, y) & \text{if } \sigma(x) = \sigma(y) = 0, \\ 2W_G(x, y) & \text{if } \sigma(x) \neq \sigma(y), \\ 0 & \text{if } \sigma(x) = \sigma(y) = 1. \end{cases}$$

Then $W_R + W_T = 4W_G$, so $R \oplus T = 4G$. By construction R has no even-even edges and T has no odd-odd edges, so both are partially independent.

We verify R is stitchable (the argument for T is identical by parity symmetry). Let x, y be odd with $W_R(x, y) = 4W_G(x, y) > 0$. Since G is stitchable, there exists even r with $W_G(x, r)W_G(r, y) \geq W_G(x, y)$. Then

$$W_R(x, r)W_R(r, y) = 2W_G(x, r) \cdot 2W_G(r, y) = 4W_G(x, r)W_G(r, y) \geq 4W_G(x, y) = W_R(x, y).$$

So R is stitchable. By Lemma 5.3, both R and T Möbius-alternate, meaning $P_R(-1)$ and $P_T(-1)$ have sign σ . Setting $A' = -A_R$ and $B' = -A_T$ (both nilpotent), we have $(I - A')^{-1} = P_R(-1)$ and $(I - B')^{-1} = P_T(-1)$ each with sign σ , and $A' + B' = -(A_R + A_T)$ is nilpotent. By Proposition 4.1,

$$(I - A' - B')^{-1} = (I + A_R + A_T)^{-1} = P_{4G}(-1) \in \text{ALT}_\sigma,$$

i.e., $4G$ Möbius-alternates. $\square$

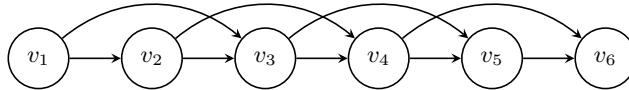
Corollary 5.5. *If G is a stitchable DAG with nonnegative edge weights, then all real roots of every walk generating polynomial $P_{G,xy}$ lie in $[-4, 0]$.*

Proof. By Corollary 5.4, $4G$ Möbius-alternates, i.e., $P_{4G}(-1) = P_G(-4)$ has sign σ . Theorem 3.2 applied with $a = 4$ gives that all real roots of $P_{G,xy}$ lie in $[-4, 0]$. $\square$

Remark 5.6 (Sharpness of the $[-4, 0]$ bound). The constant 4 in Corollary 5.5 cannot be improved for stitchable DAGs in general; whether it is sharp for chain polynomials of genuine posets is open (see below). For $n \geq 3$, let G_n be the *square of the directed path* on vertices $v_1, \dots, v_n$: unit-weight edges

$$v_i \rightarrow v_{i+1} \quad (1 \leq i \leq n-1) \quad \text{and} \quad v_i \rightarrow v_{i+2} \quad (1 \leq i \leq n-2).$$

The graph G_6 is illustrated below.



Set $\sigma(v_i) = i \bmod 2$. The only same-parity edges are the step-2 edges $v_i \rightarrow v_{i+2}$; for each such edge the intermediate vertex v_{i+1} (of opposite parity) satisfies $W(v_i, v_{i+1})W(v_{i+1}, v_{i+2}) = 1 = W(v_i, v_{i+2})$, so G_n is stitchable. Corollary 5.5 therefore guarantees that all real roots of P_{G_n, v_1, v_n} lie in $[-4, 0]$.

To find the roots explicitly, note that a walk of length k from v_1 to v_n consists of steps of size 1 or 2 summing to $n-1$; there are $\binom{k}{n-1-k}$ such walks (choosing

which $n - 1 - k$ of the k steps have size 2), giving

$$(5.7) \quad P_{G_n, v_1, v_n}(z) = \sum_{k=\lceil (n-1)/2 \rceil}^{n-1} \binom{k}{n-1-k} z^k.$$

For even $n = 2\ell$, factoring out z^ℓ yields a Chebyshev polynomial of the second kind, $U_{\ell-1}(\frac{z+2}{2})$; in general the nonzero roots of P_{G_n, v_1, v_n} are

$$(5.8) \quad -4 \cos^2\left(\frac{j\pi}{n}\right), \quad j = 1, \dots, \lfloor \frac{n-1}{2} \rfloor,$$

all of which lie strictly inside $(-4, 0)$. The most negative root $-4 \cos^2(\pi/n)$ satisfies

$$-4 \cos^2\left(\frac{\pi}{n}\right) \xrightarrow{n \rightarrow \infty} -4,$$

so no constant smaller than 4 can replace it in Corollary 5.5, and the bound $[-4, 0]$ is sharp.

The path squares G_n are not order DAGs of posets, so they leave open whether the constant improves once attention is restricted to chain polynomials. It does not: the bound $[-4, 0]$ of Proposition 6.14 is sharp, as the following family shows.

Proposition 5.7. *For each integer $k \geq 2$ let S be the adjacency matrix of the de Bruijn graph $B(k, 2)$ on the $W = k^2$ words of length two over a k -letter alphabet, and let $L_{k,T}$ be the graded poset with $\hat{0}$, $\hat{1}$, and W elements in each of the ranks $0, \dots, T$, where $(l, j) < (l+1, i)$ exactly when $S_{ij} = 1$. As $T \rightarrow \infty$ the least real root of $c_{\bar{L}_{k,T}}$ tends to the negative root of*

$$(5.9) \quad k^2 t^2 + (4k^2 - 2k)t + 1 = 0, \quad \text{which equals} \quad -4 + \frac{2}{k} + O(k^{-2}).$$

Hence no interval $[-c, 0]$ with $c < 4$ contains the real roots of the chain polynomials of all finite graded posets.

Proof. Each row and column of S has exactly k ones, and $S^2 = J$, since for words i and j there is exactly one word m with $j \rightarrow m \rightarrow i$. So $(l, j) < (l+d, i)$ holds for every i once $d \geq 2$, and is given by S when $d = 1$.

For a node v write $g_v(t) = \sum_m \#\{\text{chains of size } m \text{ with minimum } v\} t^m$. Splitting a chain at its second-smallest element gives $g_v = t(1 + \sum_{w>v} g_w)$. We claim g_v depends only on the rank of v . At rank T it is t ; and if the claim holds above rank l , with common values $h_{l+1}, h_{l+2}, \dots$, then for every lane j

$$g_{(l,j)} = t \left(1 + k h_{l+1} + W \sum_{d \geq 2} h_{l+d} \right),$$

since column j of S has k ones and all higher ranks are fully reachable. The right-hand side does not involve j , proving the claim. Writing $u_m = h_{T-m}$ and $U_m = u_0 + \dots + u_m$,

$$u_0 = t, \quad u_m = t(1 + k u_{m-1} + W U_{m-2}), \quad c_{\bar{L}_{k,T}} = 1 + W U_T.$$

Hence $F(z) = \sum_m u_m z^m$ satisfies $F(z) = t / (1 - (1 + tk)z + (tk - tW)z^2)$. For fixed t the behaviour of u_m is governed by the roots of that denominator, and as in Remark 5.10 the least root of $c_{\bar{L}_{k,T}}$ accumulates at the edge of the oscillatory regime, where the two roots collide. That is where the discriminant $(1 + tk)^2 - 4(tk - tW)$ vanishes, namely $k^2 t^2 + (4W - 2k)t + 1 = 0$; substituting $W = k^2$ gives (5.9), and

expanding its negative root in $1/k$ gives $-4 + 2/k + O(k^{-2})$. Letting $k \rightarrow \infty$ gives the last statement. $\square$

For $k = 4$ the least root is $-3.4159 \dots$ at 21 ranks, against a limit of $-3.4820 \dots$; no finite member attains -4 .

Gradedness is not a convenience. On the five-element poset with relations $0 < 2$, $0 < 3$, $0 < 4$, $1 < 3$, $1 < 4$, $2 < 3$, whose maximal chains have sizes 2 and 3, the chain polynomial is $1 + 5t + 6t^2 + t^3$ with a real root at $-5.0489 \dots$, outside $[-4, 0]$. No bound of this shape can hold without a gradedness hypothesis.

Quantitative Stitchability. Relays can pool their weight, and surplus beyond bare domination gives a constant smaller than 4.

For a same-parity pair (x, y) with $W_G(x, y) > 0$, define the *stitch ratio*

$$(5.10) \quad s(x, y) = \frac{1}{W_G(x, y)} \sum_{\substack{r: \sigma(r) \neq \sigma(x) \\ x \rightarrow r \rightarrow y}} W_G(x, r) W_G(r, y),$$

and set

$$\lambda_o = \min_{\text{odd-odd edges}} s(x, y), \quad \lambda_e = \min_{\text{even-even edges}} s(x, y),$$

with the convention $\min \emptyset = \infty$. Stitchability implies $\lambda_o, \lambda_e \geq 1$, with the relay sum witnessed by a single term; here the relays pool, and surplus ($\lambda > 1$) will be rewarded. In the unit-weight case $s(x, y)$ is simply the *number* of opposite-parity vertices r with $x \rightarrow r \rightarrow y$.

Lemma 5.8 (Quantitative form of Lemma 5.3). *Suppose G is partially independent (no even-even edges) and $\lambda_o > 0$. Then aG Möbius-alternates with respect to σ for every $a \geq 1/\lambda_o$.*

Proof. Induct on the height $h(x)$ exactly as in Lemma 5.3; write $\mu = \mu_{aG}$. The base case and Case 1 (x even) are unchanged, except that Case 1 now records, for $x \neq y$, the identity

$$(5.11) \quad |\mu(x, y)| = \sum_{x \rightarrow v} aW_G(x, v) |\mu(v, y)|.$$

In Case 2 (x odd), split out-neighbors by parity as before: $\mu(x, y) = (-1)^{\sigma(x) - \sigma(y)} (E - O)$ with

$$E = \sum_{\substack{x \rightarrow r \\ \sigma(r)=0}} aW_G(x, r) |\mu(r, y)|, \quad O = \sum_{\substack{x \rightarrow v \\ \sigma(v)=1}} aW_G(x, v) |\mu(v, y)|.$$

Every even out-neighbor $r \neq y$ satisfies (5.11); dropping the nonnegative term $r = y$ if present and grouping by v ,

$$E \geq a^2 \sum_{\sigma(v)=1} |\mu(v, y)| \sum_{\substack{\sigma(r)=0, r \neq y \\ x \rightarrow r \rightarrow v}} W_G(x, r) W_G(r, v).$$

Whenever a term matters ($W_G(x, v) > 0$ and $|\mu(v, y)| > 0$), excluding $r = y$ costs nothing: if $v = y$ the relays are even while y is odd, so $r \neq y$ automatically; if $v \neq y$, then $\mu(v, y) \neq 0$ forces a walk from v to y , so acyclicity forbids the edge $y \rightarrow v$ and the term $r = y$ vanishes from the full relay sum. For such v ,

$$\sum_{\substack{\sigma(r)=0, r \neq y \\ x \rightarrow r \rightarrow v}} W_G(x, r) W_G(r, v) = s(x, v) W_G(x, v) \geq \lambda_o W_G(x, v),$$

and therefore

$$E \geq a^2 \lambda_o \sum_{\sigma(v)=1} W_G(x, v) |\mu(v, y)| = a \lambda_o O \geq O$$

since $a \lambda_o \geq 1$. $\square$

Theorem 5.9 (Weighted stitchability). *Let G be a weighted DAG with parity function σ and $\lambda_o, \lambda_e \in (0, \infty]$. Set*

$$a^* = \left(\lambda_o^{-1/2} + \lambda_e^{-1/2} \right)^2, \quad \infty^{-1/2} := 0.$$

*If G has no same-parity edges ($\lambda_o = \lambda_e = \infty$), every real root of every $P_{G,xy}$ equals 0. Otherwise a^*G Möbius-alternates with respect to σ , and all real roots of every walk generating polynomial $P_{G,xy}$ lie in $[-a^*, 0]$.*

Proof. If there are no same-parity edges, every step of every walk changes parity, so all walks between fixed x and y have length of the same parity ε ; thus $P_{G,xy}(z) = z^\varepsilon R(z^2)$ with R having nonnegative coefficients, and the only possible real root is 0.

Otherwise put $a = a^*$ and $\alpha = \sqrt{a/\lambda_o}$. The defining identity $\sqrt{a} = \lambda_o^{-1/2} + \lambda_e^{-1/2}$ gives

$$a - \alpha = \sqrt{a} \left(\sqrt{a} - \lambda_o^{-1/2} \right) = \sqrt{a} \lambda_e^{-1/2} = \sqrt{a/\lambda_e},$$

so in particular $0 \leq \alpha \leq a$. Decompose $aG = R \oplus T$ by

$$W_R = \begin{cases} a W_G & \text{odd-odd,} \\ \alpha W_G & \text{mixed,} \\ 0 & \text{even-even,} \end{cases} \quad W_T = \begin{cases} 0 & \text{odd-odd,} \\ (a - \alpha) W_G & \text{mixed,} \\ a W_G & \text{even-even.} \end{cases}$$

Then $W_R + W_T = aW_G$, and R is partially independent. (If $\lambda_o = \infty$ then $\alpha = 0$ and R has no edges at all, so it alternates trivially; symmetrically for $\lambda_e = \infty$. Assume both finite.) For an odd-odd edge (x, y) the edge weight in R is $aW_G(x, y)$ while each relay contributes $\alpha^2 W_G(x, r) W_G(r, y)$, so

$$s_R(x, y) = \frac{\alpha^2}{a} s_G(x, y) \geq \frac{\alpha^2 \lambda_o}{a} = 1, \quad \text{since } \alpha^2 = a/\lambda_o,$$

so $\lambda_o(R) \geq 1$, and symmetrically $\lambda_e(T) \geq (a - \alpha)^2 \lambda_e / a = 1$. Lemma 5.8 (at $a = 1$, applied to each parity in turn) and Proposition 4.1 then give $P_{aG}(-1) \in \text{ALT}_\sigma$, and Theorem 3.2 finishes. $\square$

$$(\lambda_o, \lambda_e) = (1, 1) \Rightarrow a^* = 4, \quad (\lambda_o, \infty) \Rightarrow a^* = 1/\lambda_o, \quad (\lambda, \lambda) \Rightarrow a^* = 4/\lambda,$$

recovering Corollary 5.5 and Lemma 5.8 respectively, and halving the interval whenever the relay capacity doubles.

Remark 5.10 (Sharpness of a^*). The constant a^* cannot be improved as a function of (λ_o, λ_e) . For $u_o, u_e > 0$, let $Z_n(u_o, u_e)$ be the path square G_n of Remark 5.6 with unit step-1 edges, where the step-2 edge leaving v_i carries weight u_o or u_e according to the parity of v_i . Each step-2 edge has exactly one relay, of weight product 1, so $\lambda_o = 1/u_o$, $\lambda_e = 1/u_e$, and $a^* = (\sqrt{u_o} + \sqrt{u_e})^2$.

A transfer-matrix computation identifies where the roots accumulate. Fixing $z = -s$ and letting k run, $Q_k(-s) = P_{Z_n, v_1, v_k}(-s)$ is propagated by the period-2

matrices $A_u = \begin{pmatrix} -s & -su \\ 1 & 0 \end{pmatrix}$, whose product has trace $s^2 - s(u_o + u_e)$ and determinant $u_o u_e s^2$. Its eigenvalues are non-real exactly for

$$s \in \left((\sqrt{u_o} - \sqrt{u_e})^2, (\sqrt{u_o} + \sqrt{u_e})^2 \right),$$

an oscillatory band whose upper edge is precisely a^* ; in that regime $Q_n(-s)$ oscillates in n with a phase affine in s , so its sign changes $\sim n$ times across the band. For $u_o = u_e = 1$ this is the Chebyshev computation of Remark 5.6, where the accumulation is explicit. For general weights we have not proved that the roots fill the band, but exact Sturm-sequence computations support it: for $(\lambda_o, \lambda_e) = (2, 3)$ the most negative roots at $n = 24, 48, 96$ are $-1.6220, -1.6428, -1.6481$, against $a^* = 1.64983\dots$, with no root below $-a^*$ in any tested case.

The bound is also attained at the opposite end of the scale: the diamond DAG $x \rightarrow r_1, \dots, r_k \rightarrow y$ with the direct edge $x \rightarrow y$ (unit weights) has $\lambda_o = k, \lambda_e = \infty$, hence $a^* = 1/k$, and indeed $P_{x,y}(z) = z + kz^2$ has roots $\{0, -1/k\}$, attaining $-a^*$ exactly.

Remark 5.11 (Matrix formulation). With $D = \text{diag}((-1)^\sigma)$, write $DAD = S - M$, where $S \geq 0$ collects the same-parity entries of A and $M \geq 0$ the mixed ones; both are supported on edges of the DAG, and $M - S = -DAD$ is nilpotent, so the Neumann series below terminates. Möbius alternation of G is precisely entrywise nonnegativity of

$$D(I + A)^{-1}D = (I + DAD)^{-1} = \sum_{k \geq 0} (M - S)^k,$$

In matrix language the displayed condition says that $I + DAD$ is inverse-nonnegative; such matrices, and the checkerboard sign patterns producing them, are classical [BP94]. Nilpotence of DAD makes the series finite, so the spectral hypotheses usually needed there are automatic. The hypothesis of Theorem 5.9 is exactly the entrywise domination

$$(M^2)_{xy} \geq \lambda_{\sigma(x)} S_{xy}.$$

The $R \oplus T$ decomposition is intrinsically second-order; a version in which relays of length $2k$ dominate (M^{2k} -domination) is open for $k \geq 2$, and is exactly what the coprime graph of Section 8 would require.

6. CHAIN POLYNOMIALS

Recall Proposition 1.1: $c_L(t) = \frac{(t+1)^2}{t} P_{L, \hat{0}, \hat{1}}(t)$ for any finite poset L with $\hat{0} \neq \hat{1}$.

Proof of Proposition 1.1. Let $\hat{H}$ denote the collection of all chains of L (totally ordered subsets, including the empty chain). By definition,

$$(6.1) \quad c_L(t) = \sum_{C \in \hat{H}} t^{|C|},$$

and a walk of length k from $\hat{0}$ to $\hat{1}$ corresponds to a chain of size $k + 1$ containing both endpoints, so

$$(6.2) \quad P_{L, \hat{0}, \hat{1}}(t) = \sum_{\substack{C \in \hat{H}, \\ \hat{0}, \hat{1} \in C}} t^{|C|-1}.$$

Split $\hat{H}$ by which endpoints a chain contains, and let $\beta_i(t) = \sum_{C \in \beta_i} t^{|C|}$ where $\beta_0 = \{\hat{0} \in C, \hat{1} \notin C\}$, $\beta_1 = \{\hat{0} \notin C, \hat{1} \in C\}$, $\beta_2 = \{\hat{0}, \hat{1} \in C\}$, $\beta_3 = \{\hat{0}, \hat{1} \notin C\}$. Adjoining $\hat{0}$, $\hat{1}$, or both to a chain of β_3 gives bijections onto $\beta_0, \beta_1, \beta_2$ raising $|C|$ by 1, 1, 2, so

$$\beta_0 = \beta_1 = t\beta_3, \quad \beta_2 = t^2\beta_3, \quad \beta_2(t) = t P_{L, \hat{0}, \hat{1}}(t).$$

Hence $c_L = (1+t)^2\beta_3$ and $t\beta_3 = P_{L, \hat{0}, \hat{1}}$, giving $c_L(t) = \frac{(t+1)^2}{t} P_{L, \hat{0}, \hat{1}}(t)$; and $\beta_3 = c_{\bar{L}}$, which is the equivalent form. $\square$

6.1. Cohen–Macaulay Posets. Cohen–Macaulayness is sufficient for the sign hypothesis, and strictly stronger than it.

Proposition 6.1. *Let L be a finite graded poset with $\hat{0}$ and $\hat{1}$ whose order complex is Cohen–Macaulay. Then L Möbius-alternates with respect to $\sigma = \rho \bmod 2$, and all real roots of $c_L(t)$ lie in $[-1, 0]$.*

Proof. For a Cohen–Macaulay poset the order complex of each open interval (x, y) is homotopy equivalent to a wedge of spheres of dimension $\rho(y) - \rho(x) - 2$, so by Philip Hall’s theorem [Hal36; Sta11] its reduced Euler characteristic, which is $\mu_L(x, y)$, equals $(-1)^{\rho(y) - \rho(x)}$ times the number of spheres [Bjö80; Bjö82]. That is Möbius alternation, and Theorem 3.2 at $a = 1$ gives the interval, transferred to c_L by Proposition 1.1. $\square$

Geometric and distributive lattices are Cohen–Macaulay, so their $[-1, 0]$ bounds are instances.

Corollary 6.2. *For every finite geometric lattice, and for every finite distributive lattice, all real roots of $c_L(t)$ lie in $[-1, 0]$.*

Remark 6.3. In both cases the sign pattern is also available without topology. For a geometric lattice Rota’s NBC theorem [Rot64] gives $\mu_L(x, y) = (-1)^{r(y) - r(x)} B_{\text{NBC}}(x, y)$ with $B_{\text{NBC}} \geq 0$. For a distributive lattice $L \cong J(P)$ the classical Möbius formula [Rot64; Sta11] gives $\mu_L(I, J) = (-1)^{|J \setminus I|}$ when $J \setminus I$ is an antichain of P and 0 otherwise, and $|J \setminus I| = r(J) - r(I)$ in the nonzero cases. Either computation is the alternation hypothesis on the nose.

Remark 6.4. These bounds constrain the *real* roots only, so they are weaker than real-rootedness and hold unconditionally, independent of the real-rootedness conjecture for geometric lattices [AK23] and of its failure for distributive lattices [Ste07]. For Cohen–Macaulay posets the same interval follows from nonnegativity of the h -vector through the substitution $c_L(t) = (1+t)^d H\left(\frac{t}{1+t}\right)$, which sends the non-positive real roots of H into $[-1, 0]$.

6.2. Eulerian and k -Eulerian Posets. A finite graded poset L is *Eulerian* if $\mu_L(x, y) = (-1)^{\rho(y) - \rho(x)}$ on every interval [Sta94; Sta11]; face lattices of convex polytopes are the motivating example. This is verbatim the statement that L Möbius-alternates with $\sigma = \rho \bmod 2$ at $a = 1$, so Theorem 3.2 applies with no stitchability and no scaling.

Bayer and Heteyi [BH02] deform the condition by a parameter k , which moves only the scale a at which the sign pattern is read. We treat both cases together, the Eulerian one being $k = 1$.

Throughout, ρ is the rank function of L and $\rho(x, y) = \rho(y) - \rho(x)$; recall from §1 that $P_{x,y}(t) = \sum_m c_m t^m$, where c_m counts the chains $x = v_0 < v_1 < \cdots < v_m = y$ of m steps, and that $P_L(t)$ denotes the matrix $(P_{x,y}(t))_{x,y \in L}$.

Definition 6.5 ([BH02]). The k -Möbius function of a graded poset is defined recursively by

$$\mu_k([x, y]) = \begin{cases} 1 & \text{if } x = y, \\ -1 - \frac{1}{k} \sum_{x < z < y} \mu_k([x, z]) & \text{otherwise.} \end{cases}$$

A graded poset L is k -Eulerian if $\mu_k([x, y]) = (-1)^{\rho(x,y)}$ for every interval $[x, y] \subseteq L$.

At $k = 1$ the recursion is the classical one, so 1-Eulerian is Eulerian. Bayer and Hetyei show that k -Eulerian posets exist exactly for $k \in \frac{1}{2}\mathbb{Z}_{>0}$, and that every k -Eulerian poset is $2k$ -thick: each rank-2 interval has exactly $2k$ elements in its middle rank.

Lemma 6.6. For every interval $[x, y]$ of a finite graded poset with $x < y$,

$$\mu_k([x, y]) = k P_{x,y}(-1/k),$$

where $P_{x,y}$ is the walk generating polynomial of the order DAG. On the diagonal $\mu_k([x, x]) = 1$, while $k P_{x,x}(-1/k) = k$, so the identity is special to $x < y$.

By [BH02, Prop. 4.2], $\mu_k([x, y]) = -\sum_S (-1/k)^{|S|} f_S$ with f_S counting chains of (x, y) with rank set exactly S . A j -element chain has a j -element rank set and is counted by one f_S , so $\sum_{|S|=j} f_S = b_j$, the number of j -element chains of (x, y) ; and deleting endpoints gives $b_j = c_{j+1}$. With $t = -1/k$,

$$\mu_k([x, y]) = -\sum_{j \geq 0} t^j c_{j+1} = -\frac{1}{t} \sum_{m \geq 1} t^m c_m = -\frac{1}{t} P_{x,y}(t) = k P_{x,y}(t).$$

Proposition 6.7. For every finite k -Eulerian poset L , all real roots of the proper-part chain polynomial $c_{\bar{L}}(t)$ lie in $[-1/k, 0]$.

Proof. Put $\sigma = \rho \bmod 2$. For $x < y$, Lemma 6.6 turns the k -Eulerian condition $\mu_k([x, y]) = (-1)^{\rho(x,y)}$ into the statement that $P_{x,y}(-1/k)$ is nonzero of sign $(-1)^{\rho(y)-\rho(x)} = (-1)^{\sigma(x)-\sigma(y)}$. The remaining entries of the matrix $P_L(-1/k)$ are the diagonal ones, equal to 1, and those at incomparable pairs, equal to 0; both carry the required sign vacuously. All entries therefore satisfy the defining condition of ALT_σ (Definition 2.1), so $P_L(-1/k) \in \text{ALT}_\sigma$. Theorem 3.2, applied with $a = 1/k$, places all real roots of every $P_{x,y}$ in $[-1/k, 0]$, and by Proposition 1.1 the same holds for $c_{\bar{L}}$. $\square$

Corollary 6.8. For every finite Eulerian poset L , all real roots of $c_{\bar{L}}(t)$, and hence all real roots of $c_L(t)$, lie in $[-1, 0]$.

Proof. Eulerian is the case $k = 1$ of Proposition 6.7, and -1 lies in $[-1, 0]$, so the root of c_L there is harmless. $\square$

Remark 6.9. Corollary 6.8 appears to be new. For Cohen–Macaulay Eulerian posets the bound already follows from the f/h argument of §6.4, and Brenti–Welker [BW08] even give real-rootedness for barycentric subdivisions of Cohen–Macaulay complexes. But Eulerian posets need not be Cohen–Macaulay: a triangulated closed odd-dimensional manifold that is not a homology sphere, say $S^1 \times S^2$,

is Eulerian yet fails Reisner's criterion, and there the h -vector route gives nothing. Nor is real-rootedness available: Brenti and Welker conjecture it for the face lattice of every polytope, and that is open. The bound proved here is unconditional, using only the Euler relation.

Remark 6.10. It is worth comparing with the counting bound. A k -Eulerian poset is $2k$ -thick, so Corollary 6.17—if every rank-2 interval has at least c elements in its middle rank, then all real roots lie in $[-4/c, 0]$ —already gives $[-2/k, 0]$ on taking $c = 2k$; the sign argument is twice as strong. This is the same gain seen at $k = 1$, where counting gives $[-2, 0]$ and the Euler relation gives $[-1, 0]$.

The endpoint $-1/k$ is never attained. Indeed Lemma 6.6 gives $\mu_k([\hat{0}, \hat{1}]) = k P_{\hat{0}, \hat{1}}(-1/k) = -c_L(-1/k)$, and the k -Eulerian condition forces this to equal $(-1)^{\rho(\hat{0}, \hat{1})}$, so $c_L(-1/k) = \pm 1 \neq 0$.

The interval is nevertheless the right one, by a thickening operation of Bayer and Hetyei. For $\ell \geq 1$ let $D_\ell L$ be obtained from L by replacing each element of $\bar{L}$ with ℓ pairwise incomparable copies, setting $x_i < y_j$ if and only if $x < y$. Then $f_S(D_\ell L) = \ell^{|S|} f_S(L)$ for every set S of ranks, so

$$(6.3) \quad c_{D_\ell L}(t) = c_L(\ell t) :$$

thickening rescales the variable, dividing every root by ℓ and preserving real-rootedness. Since L is k -Eulerian if and only if $D_\ell L$ is $k\ell$ -Eulerian [BH02, Cor. 4.5], thickening an Eulerian poset produces ℓ -Eulerian posets whose roots are those of L scaled by $1/\ell$.

The Boolean lattice B_n supplies the extremal base. Chains in $\bar{B}_n$ are ordered set partitions, so $c_{\bar{B}_n}(t) = \sum_j (j+1)! S(n, j+1) t^j$ with $S(\cdot, \cdot)$ the Stirling numbers of the second kind, and its most negative root is $-0.95868, -0.99564, -0.99975, -0.99998$ for $n = 5, 8, 12, 16$. Proposition 6.11 below shows that these tend to -1 , from inside $(-1, 0)$ and at an exponential rate. By (6.3) the most negative root of $c_{D_\ell \bar{B}_n}$ then tends to $-1/\ell$, so no interval smaller than $[-1/k, 0]$ suffices. By contrast the rank-uniform poset with ℓ elements in each interior rank is D_ℓ applied to a chain, and a chain of rank $n+1$ is only half-Eulerian with $c_L(t) = (1+t)^n$; its root at -1 sits at half of the bound $[-2, 0]$ that $k = 1/2$ allows, and that factor of two is inherited by all its thickenings. At $k = 1$ the extremal family is the Boolean lattice itself, so $[-1, 0]$ is already asymptotically sharp for Eulerian posets.

Proposition 6.11 (Sharpness of $[-1/k, 0]$). *Let $n \geq 2$ and let $t_{\min}(n)$ be the most negative root of $c_{\bar{B}_n}$. All roots of $c_{\bar{B}_n}$ are real and lie in $(-1, 0)$, and*

$$(6.4) \quad t_{\min}(n) \leq -\frac{2^n - n - 1}{2^n - 2} \xrightarrow{n \rightarrow \infty} -1.$$

Consequently, for every integer $k \geq 1$ the k -Eulerian posets $D_k B_n$ satisfy $t_{\min}(D_k B_n) = t_{\min}(n)/k \rightarrow -1/k$, so for such k the interval $[-1/k, 0]$ of Proposition 6.7 cannot be replaced by any smaller one—though its endpoint is attained by no k -Eulerian poset. The thickening D_k produces k copies of each element, so it realises only integer k ; for the half-integer values $k \in \frac{1}{2} + \mathbb{Z}_{\geq 0}$ also permitted by the definition, sharpness is open.

Proof. The order complex of $\bar{B}_n$ is the barycentric subdivision $\text{sd } \partial \Delta^{n-1}$ of the boundary of the $(n-1)$ -simplex, a simplicial sphere of dimension $d-1$ with $d = n-1$, whose f -polynomial is $c_{\bar{B}_n}$. Writing $H(s) = \sum_i h_i s^i$ for its h -polynomial, the two

are related by the substitution of §6.4,

$$(6.5) \quad c_{\bar{B}_n}(t) = (1+t)^d H\left(\frac{t}{1+t}\right).$$

Since $\partial\Delta^{n-1}$ has h -vector $(1, 1, \dots, 1)$, in particular nonnegative, H is real-rooted by Brenti–Welker [BW08]; and $\bar{B}_n$ is shellable, so $\text{sd } \partial\Delta^{n-1}$ is Cohen–Macaulay and the coefficients of H are nonnegative. As $h_0 = 1 \neq 0$, the roots of H are therefore strictly negative, say $-r_1, \dots, -r_d$ with $r_i > 0$. Being the h -vector of a sphere it satisfies $h_0 = h_d = 1$ and, by Dehn–Sommerville, $h_{d-1} = h_1 = f_0 - d$, where $f_0 = |\bar{B}_n| = 2^n - 2$ counts the vertices. Hence H is monic of degree d and

$$\prod_i r_i = h_0 = 1, \quad \sum_i r_i = h_{d-1} = (2^n - 2) - (n - 1) = 2^n - n - 1.$$

By (6.5) the increasing bijection $s \mapsto t = s/(1-s)$ from $(-\infty, 0)$ onto $(-1, 0)$ carries the roots of H onto those of $c_{\bar{B}_n}$, sending $-r$ to $-r/(1+r)$; the latter are thus real and contained in $(-1, 0)$. Since $r \mapsto r/(1+r)$ is increasing and $\max_i r_i \geq (\sum_i r_i)/d = (2^n - n - 1)/(n - 1)$,

$$t_{\min}(n) = -\frac{\max_i r_i}{1 + \max_i r_i} \leq -\frac{2^n - n - 1}{(2^n - n - 1) + (n - 1)} = -\frac{2^n - n - 1}{2^n - 2},$$

which is (6.4). Finally B_n is Eulerian, so $D_k B_n$ is k -Eulerian by [BH02, Cor. 4.5], and (6.3) divides each of its roots by k ; the endpoint $-1/k$ is not attained because $c_{\bar{L}}(-1/k) = \pm 1$ for every k -Eulerian L , as shown above. $\square$

Remark 6.12. Only the rate (6.4) is new. That $c_{\bar{B}_n}$ is real-rooted is classical— B_n is shellable, so the h -vector of its order complex is nonnegative and Brenti–Welker applies—but real-rootedness alone says nothing about *where* the extreme root sits, and it is exactly that location which the sharpness statement needs.

6.3. Graded Posets. Graded posets need not be Cohen–Macaulay, and real-rootedness is delicate here, failing already for the W -polynomial of labeled posets [Brä04; Ste07]; the $[-4, 0]$ bound on the real roots nonetheless holds unconditionally.

Proposition 6.13. *For every finite graded poset L (with rank function ρ , minimum $\hat{0}$, and maximum $\hat{1}$), the DAG $4L$ Möbius-alternates with $\sigma(x) = \rho(x) \bmod 2$.*

Proof. Let $x < y$ with $\sigma(x) = \sigma(y)$, so $\rho(y) - \rho(x)$ is even and positive, hence ≥ 2 . A maximal chain $x = v_0 < v_1 < \dots < v_m = y$ has $m = \rho(y) - \rho(x) \geq 2$, so $x < v_1 < y$ with $\sigma(v_1) \neq \sigma(x)$: a stitch for (x, y) . Now apply Corollary 5.4. $\square$

Proposition 6.14. *For every finite graded poset L with a unique minimum $\hat{0}$ and maximum $\hat{1}$, all real roots of $c_L(t)$ lie in $[-4, 0]$.*

Proof. By Proposition 1.1, $c_L(t) = \frac{(t+1)^2}{t} P_{L, \hat{0}, \hat{1}}(t)$. Proposition 6.13 gives that $4L$ Möbius-alternates, and Corollary 5.5 gives that all real roots of $P_{L, \hat{0}, \hat{1}}(t)$ lie in $[-4, 0]$. Hence so do those of $c_L(t)$. $\square$

The minimum and maximum are not actually needed: they can be adjoined for free.

Corollary 6.15. *Let P be any finite graded poset (every maximal chain of P has the same length), not necessarily possessing a minimum or maximum. Then all real roots of $c_P(t)$ lie in $[-4, 0]$.*

Proof. Adjoin a new global minimum and maximum: $\hat{P} = P \cup \{\hat{0}, \hat{1}\}$. Every maximal chain of $\hat{P}$ contains $\hat{0}$ and $\hat{1}$, and its interior is a maximal chain of P ; since all of the latter share one length, so do all of the former, and $\hat{P}$ is a graded poset with minimum and maximum. Every chain of $\hat{P}$ is a chain of P together with a subset of $\{\hat{0}, \hat{1}\}$, so

$$c_{\hat{P}}(t) = (1+t)^2 c_P(t).$$

By Proposition 6.14 all real roots of $c_{\hat{P}}(t)$ lie in $[-4, 0]$; dividing by $(1+t)^2$ removes only roots at $-1 \in [-4, 0]$, so all real roots of $c_P(t)$ lie in $[-4, 0]$ as well. $\square$

Remark 6.16. For geometric and distributive lattices the stronger $[-1, 0]$ bound holds (Corollary 6.2); the factor of 4 here comes from the $4G$ scaling in Corollary 5.4. How much of it is necessary is quantified in Remark 5.6: the optimal constant for graded posets lies in $[3.3745, 4]$.

With unit weights the stitch ratio $s(x, y)$ counts opposite-parity elements strictly between x and y , so counting intermediates improves the constant.

Corollary 6.17. *Let L be a finite graded poset with $\hat{0}$ and $\hat{1}$ in which every rank-2 interval has at least $c \geq 1$ elements in its middle rank. Then all real roots of the proper-part chain polynomial $c_{\bar{L}}(t)$ lie in $[-4/c, 0]$; equivalently, all real roots of $c_L(t)$ other than the forced root at $t = -1$ lie there. (For $c \leq 4$ the point -1 lies in the interval, so in that range all real roots of $c_L(t)$ itself lie in $[-4/c, 0]$.)*

Proof. Work in the order DAG of L with $\sigma(x) = \rho(x) \bmod 2$, as in Proposition 6.13. Let $x < y$ with $\sigma(x) = \sigma(y)$, so $\rho(y) - \rho(x) \geq 2$. Take a maximal chain $x = v_0 < v_1 < v_2 < \dots < y$ and set $z = v_2$, so $[x, z]$ is a rank-2 interval with $z \leq y$. Each of its ($\geq c$) middle-rank elements r satisfies $x < r < y$ and $\sigma(r) \neq \sigma(x)$, contributing 1 to the relay sum in the order DAG. Hence $s(x, y) \geq c$ for every same-parity pair, so $\lambda_o, \lambda_e \geq c$ and Theorem 5.9 places all real roots of $P_{L, \hat{0}, \hat{1}}$ in $[-4/c, 0]$. Since $P_{L, \hat{0}, \hat{1}}(t) = t c_{\bar{L}}(t)$ (Proposition 1.1), the same holds for $c_{\bar{L}}(t)$, and $c_L(t) = (1+t)^2 c_{\bar{L}}(t)$ gives the statement for c_L . $\square$

In subspace lattices every rank-2 interval has $q+1$ middle elements, and known real-rootedness upgrades the conclusion to all roots.

Theorem 6.18 (Subspace lattices). *For the lattice $L_n(\mathbb{F}_q)$ of subspaces of $\mathbb{F}_q^n$, all roots of the proper-part chain polynomial $c_{\bar{L}_n(\mathbb{F}_q)}(t)$ —real and complex—lie in the interval $[-4/(q+1), 0]$. Equivalently, apart from the root at $t = -1$, all roots of $c_{L_n(\mathbb{F}_q)}(t)$ lie there.*

Proof. Every rank-2 interval of $L_n(\mathbb{F}_q)$ is isomorphic to $L_2(\mathbb{F}_q)$, whose middle rank consists of the $q+1$ lines of $\mathbb{F}_q^2$; Corollary 6.17 with $c = q+1$ places all real roots of $c_{\bar{L}}(t)$ in $[-4/(q+1), 0]$. The chain polynomial of $L_n(\mathbb{F}_q)$ is real-rooted by Athanasiadis–Kalampoglia–Evangelinou [AK23, Thm. 1.3 and Prop. 3.1]; more generally subspace lattices are perfect matroid designs (any two flats of equal rank have the same number of atoms below them), for which real-rootedness is due to Brändén–Saud Maia Leite [BS25]. Since c_L and $c_{\bar{L}}$ differ by the factor $(1+t)^2$, both are then real-rooted. All roots of $c_{\bar{L}}$ are therefore real, and confined to $[-4/(q+1), 0]$. For c_L this adds a root at -1 of multiplicity at least two, and exactly two when $q \geq 4$ (since $c_L(t)/(1+t)^2 = c_{\bar{L}}(t)$ is nonvanishing at $t = -1$, its roots lying in $[-4/(q+1), 0]$ with $4/(q+1) < 1$ for $q \geq 4$; for $q \leq 3$ the interval contains -1 in any case). $\square$

Remark 6.19. For $q \geq 4$ the interval is a proper subinterval of $[-1, 0]$, below the reach of the h -vector route. Since all roots of $c_{\bar{L}}$ are confined, the chain polynomial is controlled everywhere by its leading coefficient alone: writing $c_{\bar{L}}(t) = N \prod_{i=1}^{n-1} (t - s_i)$ with each $s_i \in [-4/(q+1), 0]$ and N the number of complete flags, every evaluation satisfies $|c_{\bar{L}}(t)| \leq |1+t|^2 N \left(|t| + \frac{4}{q+1}\right)^{n-1}$. Localization on an interval shrinking with q appears to be new for subspace lattices. We have verified the bound numerically for $n \leq 5, q \leq 7$.

Remark 6.20. Nothing in the argument is special to subspace lattices. A geometric lattice is a *perfect matroid design* if any two flats of the same rank have the same number of atoms below them; the real-rootedness theorem of [BS25] covers this whole class. Consequently, if every rank-2 interval of such a lattice has at least c elements in its middle rank, all roots of $c_{\bar{L}}$ lie in $[-4/c, 0]$. This improves on the geometric-lattice bound $[-1, 0]$ of Corollary 6.2 exactly when $c \geq 5$. For the affine geometry $AG(n, q)$ the rank-2 intervals have q middle elements, giving $[-4/q, 0]$ once $q \geq 5$.

Remark 6.21 (Root location and vertex percolation). The interval $[-a, 0]$ has a probabilistic reading, which indicates why the *location* of the roots carries information that real-rootedness alone does not. Write $\Delta(Q)$ for the order complex of a poset Q , the simplicial complex whose faces are the chains of Q , and recall that $c_0 = 1$ counts the empty chain, matching the convention $\tilde{\chi}(\emptyset) = -1$ for the reduced Euler characteristic. Retain each element of $\bar{L}$ independently with probability p and let $\bar{L}_p$ be the resulting random induced subposet. A k -element chain survives precisely when all k of its elements are retained, so linearity of expectation gives

$$(6.6) \quad \mathbb{E}[\tilde{\chi}(\Delta(\bar{L}_p))] = \sum_{k \geq 0} (-1)^{k-1} p^k c_k(\bar{L}) = -c_{\bar{L}}(-p),$$

The roots of $c_{\bar{L}}$ lying in $[-1, 0]$ are therefore exactly the densities p at which the expected reduced Euler characteristic vanishes.

Since p is a probability, the transition statement below needs $a < 1$; for $a \geq 1$, in particular for $[-4, 0]$, it is empty. Confinement still constrains the curve: if the roots are real and lie in $[-a, 0]$, writing $c_{\bar{L}}(t) = N \prod_i (t + r_i)$ with $r_i \in [0, a]$ and N the number of maximal chains gives $|\mathbb{E}[\tilde{\chi}(\Delta(\bar{L}_p))]| \leq N a^{\deg c_{\bar{L}}}$ for all $p \in [0, 1]$.

When $a < 1$ the bound localizes a transition as well. Indeed $c_{\bar{L}}$ has no root in $[-1, -a)$, so the expectation (6.6), being continuous in p , cannot change sign on $(a, 1]$; at $p = 1$ it equals $\tilde{\chi}(\Delta(\bar{L})) = \mu_L(\hat{0}, \hat{1})$ by Philip Hall's theorem. Its sign is thus constant and known on all of $(a, 1]$, and any change of sign occurs at density at most a . For $L_n(\mathbb{F}_q)$ with $q \geq 4$, Theorem 6.18 confines the transition to densities below $4/(q+1)$.

The identity (6.6) is elementary and concerns the expectation only; the variance of $\tilde{\chi}(\Delta(\bar{L}_p))$ is not a function of $c_{\bar{L}}$, and no concentration statement is implied. Zeros of expected Euler characteristic curves have been studied by Bobrowski and Skraba [BS20], who give experimental evidence that in geometric models they approximate homological percolation thresholds.

6.4. Barycentric Subdivisions. Brenti and Welker [BW08, Thm. 3.1] prove that the h -polynomial of a barycentric subdivision $\text{sd } \Delta$ has only real and simple zeros whenever Δ is a Boolean cell complex—in particular whenever it is simplicial—with a nonnegative h -vector. For pure complexes with negative h -entries we are

not aware of any previous statement on the location of these roots. Delucchi, Pixton and Sabalka [DPS12] treat iterated subdivisions instead: the roots of $f_{\text{sd}^m \Delta}$ converge as $m \rightarrow \infty$ to limits depending only on the dimension, and are symmetric about -2 . What follows concerns a single subdivision of a fixed complex.

Write $f_\Delta(t) = \sum_{i \geq 0} f_{i-1}(\Delta) t^i$ for the f -polynomial, where f_{i-1} counts the i -element faces and $f_{-1} = 1$ counts the empty face. Chains of faces of Δ are exactly the faces of its barycentric subdivision $\text{sd } \Delta$, so the chain polynomial of the face poset of Δ is $f_{\text{sd } \Delta}$. The face poset is graded precisely when Δ is pure, so Corollary 6.15 applies with no further hypothesis.

Corollary 6.22. *For every pure simplicial complex Δ , all real roots of the f -polynomial of $\text{sd } \Delta$ lie in $[-4, 0]$. If in addition every ridge of Δ lies in at least two facets—for instance if Δ is a closed pseudomanifold—then all real roots lie in $[-2, 0]$.*

Proof. The first claim is Corollary 6.15 applied to the face poset, whose elements we identify with their vertex sets, ranked by cardinality. For the second we apply Corollary 6.17, which requires a lower bound c on the number of middle-rank elements of every rank-2 interval; adjoining $\hat{0}$ and $\hat{1}$ leaves $c_{\bar{L}}$ unchanged and lets the count be made uniformly. An interval $[F, G]$ with $|G| = |F| + 2$ has the two sets $F \cup \{v\}$, $v \in G \setminus F$, in its middle rank, and both are faces; $[\hat{0}, e]$ for an edge e has the two vertices of e ; and $[R, \hat{1}]$ for a ridge R has the facets containing R , at least two by hypothesis. So $c = 2$, and Corollary 6.17 gives $[-4/2, 0]$. $\square$

Remark 6.23. The window $(-2, -1)$ is genuinely occupied. Let Δ be pure of even dimension d with every ridge in at least two facets and $\chi(\Delta) \leq 0$. The f -polynomial of $\text{sd } \Delta$ has degree $d + 1$, which is odd, and positive leading coefficient, so it tends to $-\infty$ as $t \rightarrow -\infty$; at $t = -1$ it equals $1 - \chi(\Delta) \geq 1 > 0$. Hence it has a real root below -1 , necessarily in $[-2, -1)$ by Corollary 6.22. For the 7-vertex torus, with f -vector $(7, 21, 14)$ and $\chi = 0$, the f -polynomial of the subdivision is $1 + 42t + 126t^2 + 84t^3$, whose most negative real root is $-1.0223\dots$

7. AN APPLICATION: NORMAL-SUBGROUP CHAINS

Normality is not transitive— $H \trianglelefteq K \trianglelefteq G$ does not force $H \trianglelefteq G$ —so the normal-subgroup relation is a DAG but not the order DAG of any poset, and the classical incidence algebra is unavailable ([Fin70, Lemma 2.3]). The present method works in the full matrix algebra and is untouched by the distinction.

Fix a finite group G_0 , and let T be the DAG whose vertices are the subgroups of G_0 , with a directed edge $G \rightarrow H$ (weight 1) whenever H is a proper normal subgroup of G . A walk from G_0 to H_0 in T is precisely a chain $G_0 \triangleright G_1 \triangleright \dots \triangleright H_0$ of iterated proper normal subgroups.

Proposition 7.1. *For every finite group G_0 and every subgroup $H_0 \leq G_0$, all real roots of $P_{T, G_0, H_0}(t)$ lie in $[-4, 0]$.*

Proof. Take $\sigma(G) = \text{len}(G) \bmod 2$, the composition length mod 2. Let $H \triangleleft G$ be proper with $\sigma(H) = \sigma(G)$; then $\text{len}(G) - \text{len}(H)$ is positive and even, hence ≥ 2 . By Jordan–Hölder [DF04] take a composition series $H = C_0 \trianglelefteq \dots \trianglelefteq C_n = G$ and put $r = C_{n-1}$:

$$r \trianglelefteq G, \quad H \trianglelefteq r \text{ (as } xHx^{-1} = H \text{ for } x \in r \subseteq G), \quad \text{len}(r) = \text{len}(G) - 1 > \text{len}(H),$$

so $r \neq H$ and $\sigma(r) \neq \sigma(H)$. Thus T is stitchable, and Corollaries 5.4 and 5.5 give the bound. $\square$

Since G_0 was arbitrary, a single interval bounds the roots between *any* two finite groups in the normal-subgroup relation. (If H_0 is not subnormal in G_0 there is no such chain and $P_{T,G_0,H_0} \equiv 0$; as everywhere in this paper, root-localization statements are read as claims about the nonzero walk polynomials.)

8. OPEN PROBLEM: THE COPRIME GRAPH

The divisibility relation yields at once: its intervals are distributive lattices, so the classical Möbius formula supplies the alternating signs and all its walk generating polynomials have real roots in $[-1, 0]$. The coprime relation does not.

Let G be the DAG of integers under the coprime-less-than relation:

$$x \rightarrow y \iff y < x \text{ and } \gcd(x, y) = 1.$$

This graph is infinite, so $(I+A)^{-1}$ is not defined in the finite-dimensional sense used above. It is however locally finite with finite intervals: only finitely many walks join a fixed pair $x > y$, so $P_{G,x,y}$ and $\mu_G(x, y) = P_{G,x,y}(-1)$ are given by finite sums, and every statement below is read entrywise in that sense. Equivalently one may work in the finite truncations $G_N = G[\{0, 1, \dots, N\}]$, on which $P_{G_N,x,y} = P_{G,x,y}$ for all $N \geq x$. Computer evidence strongly suggests that G Möbius-alternates, but no proof is known, not even that kG alternates for any specific k . If G does alternate, the parity function must be $\sigma(x) = x \bmod 2$, since the longest path from x to $y < x$ is $x \rightarrow x-1 \rightarrow x-2 \rightarrow \dots \rightarrow y$.

The Möbius function of G carries arithmetic content: with ϕ the Euler totient and the convention $\phi(0) = 1$, for all $n \neq 1$,

$$\sum_{0 \leq t \leq n} \phi(t) \mu_G(n, t) = 1.$$

The graph G is partially independent but not stitchable. However, data suggests the failure is minimal. Call a same-parity edge *unstitchable* if its endpoints are coprime with no even coprime integer strictly between them. A search over all such pairs with $x \leq 6 \cdot 10^4$ and no restriction on $x - y$ ($3.6 \cdot 10^8$ pairs) found four unstitchable pairs, every one of gap 16; a search over all pairs with $x \leq 10^8$ and $x - y \leq 256$ ($5.2 \cdot 10^9$ pairs) found 6660, again all of gap 16, every one lying in one of the two arithmetic progressions

$$(12831+30030k, 12815+30030k) \quad \text{and} \quad (17215+30030k, 17199+30030k), \quad k \geq 0,$$

where $30030 = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$. Within these ranges the two progressions are therefore exactly the unstitchable pairs; that they exhaust all of them is open.

A proof would require: (i) confirming that these two families exhaust all unstitchable pairs of G (open); (ii) verifying that G Möbius-alternates on each such pair directly; and (iii) showing that non-nested unstitchable pairs do not obstruct global alternation. Progress has been made on steps (ii) and (iii); step (i) remains open.

In the language of Theorem 5.9 the obstruction is stark: the unstitchable pairs have stitch ratio 0, so no second-order criterion applies, however quantitative. What is missing is exactly the higher-order (M^{2k} -domination) version flagged in Remark 5.11.

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